

PRAIRIE DU CHIEN, WI, USA

WMO: 726444

Lat: 43.019N

Lon: 91.124W

Elev: 661

StdP: 14.35

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04963

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.0	-2.5	-16.4	2.3	-7.0	-11.5	3.0	-1.6	24.7	18.1	21.3	21.4	3.4	280	0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	90.4	74.9	87.4	72.1	84.2	71.0	77.3	86.3	75.2	83.7	73.5	81.6	8.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.6	132.6	82.6	72.5	123.6	80.4	70.9	116.7	78.8	41.1	85.8	39.1	83.8	37.5	81.9	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.4	(b) 17.9	(c) 15.8	(e) -15.6	(f) 94.2	(g) 5.6	(h) 2.6	(i) -19.6	(j) 96.1	(k) -22.9	(l) 97.6	(m) -26.0	(n) 99.1	(o) -30.1	(p) 101.0		
(a) 20.4	(b) 17.9	(c) 15.8	(e) -16.1	(f) 80.4	(g) 5.6	(h) 2.7	(i) -20.1	(j) 82.4	(k) -23.3	(l) 83.9	(m) -26.4	(n) 85.5	(o) -30.5	(p) 87.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	48.6	21.0	23.0	36.1	48.7	59.6	69.1	73.0	70.7	64.1	51.1	38.8	26.3	(6)
(7)		DBStd	20.63	12.75	11.92	11.95	9.67	8.59	6.09	5.68	5.47	7.72	9.19	10.08	11.12	(7)
(8)		HDD50	3457	900	755	456	137	16	0	0	0	3	97	356	737	(8)
(9)		HDD65	6725	1365	1175	899	495	213	27	6	11	107	441	785	1201	(9)
(10)		CDD50	2944	0	0	24	98	313	573	714	641	427	132	21	1	(10)
(11)		CDD65	737	0	0	2	7	45	150	255	186	81	11	0	0	(11)
(12)		CDH74	6517	0	0	16	89	448	1342	2301	1526	700	94	1	0	(12)
(13)		CDH80	1937	0	0	1	17	118	394	779	429	183	16	0	0	(13)
(14)	Wind	WSAvg	5.8	6.3	6.3	6.7	7.6	6.4	5.1	4.2	3.9	4.7	5.9	6.3	6.1	(14)
(15)	Precipitation	PrecAvg	31.9	1.0	1.1	1.9	3.3	4.0	4.1	3.5	4.1	3.2	2.2	2.1	1.3	(15)
(16)		PrecMax	47.8	3.0	3.3	5.2	6.7	11.7	10.0	7.3	10.2	7.8	5.6	5.9	3.0	(16)
(17)		PrecMin	20.4	0.1	0.0	0.0	1.0	0.8	0.6	0.6	0.4	0.1	0.0	0.1	0.2	(17)
(18)		PrecStd	6.2	0.6	0.7	1.2	1.5	1.9	2.5	1.8	2.4	2.1	1.2	1.4	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	56.7	76.7	82.2	89.7	91.9	95.0	92.8	90.2	82.1	70.7	57.3	(19)
(20)			MCWB	42.6	47.9	62.1	63.3	72.1	74.5	77.9	75.7	72.8	67.2	57.6	53.2	(20)
(21)		2%	DB	45.3	48.4	65.6	76.1	82.6	88.4	90.4	88.1	84.8	76.1	63.8	48.8	(21)
(22)			MCWB	39.8	42.8	56.2	60.0	65.5	71.9	76.3	73.3	70.7	63.1	54.5	44.6	(22)
(23)		5%	DB	39.8	43.0	59.5	70.5	79.1	84.2	87.2	84.3	81.5	71.8	59.1	44.7	(23)
(24)			MCWB	35.7	38.1	51.5	56.0	63.9	69.7	73.4	71.6	67.9	61.7	51.0	40.8	(24)
(25)		10%	DB	36.8	38.6	53.6	65.7	75.0	81.5	84.0	81.8	77.9	66.4	54.5	40.5	(25)
(26)			MCWB	34.0	34.6	46.2	52.9	62.0	68.4	72.2	70.3	65.9	57.9	47.3	36.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.1	48.6	63.7	66.4	73.5	77.8	81.5	78.8	75.7	71.1	59.9	55.0	(27)
(28)			MCDWB	50.3	56.5	76.0	77.8	83.4	87.2	90.4	88.4	84.9	79.0	67.1	58.5	(28)
(29)		2%	WB	39.8	43.1	58.7	61.5	69.4	75.1	78.1	76.1	72.8	66.9	56.3	45.6	(29)
(30)			MCDWB	44.7	47.6	64.3	72.4	78.7	84.6	86.3	84.2	81.4	73.0	62.4	47.2	(30)
(31)		5%	WB	36.5	38.3	51.4	58.4	66.8	72.9	75.8	74.4	70.7	62.9	52.4	40.7	(31)
(32)			MCDWB	39.6	42.5	58.3	68.1	75.2	82.1	84.6	81.2	78.1	69.4	58.0	44.2	(32)
(33)		10%	WB	34.2	35.1	46.3	54.6	64.2	70.6	73.9	72.5	68.5	58.7	48.1	37.2	(33)
(34)			MCDWB	36.8	38.2	52.9	64.3	72.5	78.4	82.3	79.3	75.3	65.8	54.1	40.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.3	16.5	17.1	19.6	20.2	19.1	18.7	18.9	20.5	18.9	15.9	14.0	(35)
(36)			MCDBR	17.5	20.2	24.2	27.5	24.5	22.7	21.1	21.4	23.2	24.5	22.8	18.1	(36)
(37)			MCWBR	13.3	15.4	15.0	14.8	11.4	10.5	9.5	9.8	10.9	13.8	15.2	14.6	(37)
(38)		5% WB	MCDWB	16.2	19.6	22.6	23.0	20.6	19.4	18.8	17.9	19.0	20.7	20.9	17.0	(38)
(39)			MCWBR	12.7	15.3	15.0	14.4	11.5	11.2	9.9	9.9	10.5	14.4	15.8	14.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.277	0.301	0.331	0.375	0.403	0.425	0.434	0.417	0.385	0.338	0.297	0.278	(40)	
(41)		taud	2.381	2.320	2.377	2.297	2.272	2.249	2.226	2.275	2.342	2.471	2.507	2.430	(41)	
(42)		Ebn at Noon	274	284	289	283	276	269	266	266	267	269	266	264	(42)	
(43)		Edh at Noon	25	31	34	40	42	43	43	40	35	27	22	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	567	893	1208	1452	1723	1912	1944	1681	1381	892	592	434	(44)	
(45)		RadStd	61	90	135	126	126	152	110	114	108	115	46	40	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-7.13	N/A	N/A	-1.26	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page